WATER PURIFICATION

(DIFFERENT TYPES OF WATER IDENTIFIER – ML & AI)

INDIVIDUAL THESIS

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DECLARATION AND SUPERVISOR CERTIFICATION

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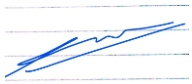


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Supervisor certification. The above candidate carried out research for the bachelor's degree thesis under my supervision.

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ABSTRACT

Reliable identification of water types from low-cost physicochemical sensor data may support rapid screening and improved decision-making in intelligent water-management systems. This individual research component investigates a proposed machine-learning-based classifier for distinguishing distilled, tap, mineral, alkaline and potentially contaminated water. The proposal specifies pH, total dissolved solids, turbidity, electrical conductivity and temperature as candidate inputs collected through a microcontroller-connected sensor array. The research design combines controlled sampling and calibration, sample-aware data partitioning, preprocessing pipelines and comparative evaluation of conventional classifiers, including logistic regression, k-nearest neighbours, support vector machines, random forests and gradient boosting. An architectural design links the sensor interface to an inference service, an audit log and a user-facing display. Research validity is addressed through independent-sample testing, class-level error analysis, latency measurement, sensor-drift checks and explicit abstention when measurements are insufficient. This document consolidates the original proposal and specifies an evidence-based validation plan; it does not claim that training, deployment or experimental performance results have been independently verified. Quantitative outcomes and final conclusions must be updated after the author provides implementation artefacts, recorded measurements and approved test results. The intended contribution is a reproducible workflow for water-type identification rather than a certification of drinking-water safety.

Keywords: water-type classification; machine learning; IoT sensing; physicochemical parameters; experimental validation

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LIST OF ABBREVIATIONS

Table P.1: Alphabetical abbreviations

Abbreviation	Meaning
ADC	Analog-to-digital converter
AI	Artificial intelligence
ANN	Artificial neural network
API	Application programming interface
CSV	Comma-separated values
EC	Electrical conductivity
ESP32	Espressif 32-bit microcontroller family
F1	Harmonic mean of precision and recall
IoT	Internet of Things
KNN	K-nearest neighbours
ML	Machine learning
NTU	Nephelometric turbidity unit
pH	Measure of hydrogen ion activity
RF	Random forest
SVM	Support vector machine
TDS	Total dissolved solids
WHO	World Health Organization

1 INTRODUCTION

1.1 Overall Project Context

Secure access to reliable water information depends on both measuring water characteristics and interpreting those measurements in a form that supports appropriate decisions. Chemical and microbiological assessments remain essential where safety is at issue, but laboratory turnaround time, equipment requirements and the need for trained personnel limit their suitability for frequent low-resource screening. The World Health Organization (WHO) emphasises risk-based drinking-water management and independent surveillance instead of treating one quick instrumental reading as a complete safety test [1].

The group research project, Water Purification [11], investigates a broader intelligent water-management workflow. Within this project, the individual component documented here is identified in the original proposal as “Different Types of Water Identifier – ML & AI”. Its purpose is narrower than the overall purification platform: it aims to assign a water-type category from measured physicochemical parameters. An assigned category may inform subsequent monitoring or treatment decisions; it cannot by itself establish potability or identify every chemical and biological hazard.

The proposal identifies a possible sensor array comprising pH, total dissolved solids (TDS), turbidity, conductivity and temperature, together with an ESP32 or comparable microcontroller, data-processing software and a user dashboard. These are design inputs from the proposal, not verified descriptions of a completed prototype. The final system boundary must be checked against the author’s latest hardware and software artefacts before submission. This distinction supports an accurate research narrative and prevents planned functionality from being reported as an achieved result.

1.2 Individual Component Context

Water types such as distilled, tap, alkaline and mineral water can differ in dissolved-ion concentration, ionic composition, turbidity and acidity. However, their sensor signatures can overlap, and a water-type name is not automatically an objective ground-truth label. The source, manufacturing process, sample handling and independent reference measurement should therefore be retained with every labelled sample. A defensible classifier must learn from independently characterised samples rather than merely encode arbitrary concentration cut-offs.

In a multi-component team project, water classification may provide an upstream signal to a purification or advisory module. The individual contribution should remain traceable through a defined interface: a structured sensor packet enters the identification service and a predicted class, confidence or uncertainty indicator, data-quality flags, model version and timestamp leave it. Decisions about treatment, dosing or consumption belong to other appropriate safeguards and components, not automatically to this classifier.

Table 1.1: Project and individual component boundary

Element	Within individual research	Outside direct claim
Sensor stream	Acquisition schema, preprocessing, data-quality control	Independent laboratory certification
Model	Water-type prediction and evaluation	Microbiological pathogen detection
Interface	Prediction, evidence, timestamp and flags	Unsupervised potable-water guarantee
Integration	Documented input/output contracts	Automated treatment proved effective

1.3 Problem Statement

Conventional low-cost monitors frequently provide numeric sensor readings or simple threshold-based warnings, but neither a single threshold nor a bare measurement directly answers the categorical question “what kind of water is this sample?”. Conversely, machine-learning demonstrations based only on stored tables may understate the problems introduced by field measurements, sensor drift, variable temperatures, dependent repeated readings and inconsistent sample labels. The core engineering and research problem is how to link a calibrated multi-sensor stream to a validated, transparent, reproducible water-type classification result.

An additional difficulty is the semantic difference between water type and water safety. A classifier might identify a sample as tap water although it contains an unmeasured hazard. Likewise, “contaminated” is a broad class that cannot be identified with certainty from pH, TDS, conductivity, turbidity and temperature alone. The operational system must therefore declare its limits, expose uncertainty and avoid presenting predicted class labels as public-health clearance.

1.4 Research Gap

The original proposal frames the gap in three dimensions: a conceptual shift from binary anomaly alerts toward categorical identification, a systemic need to connect live sensor data to practical machine learning, and an actionability requirement linking predictions to understandable outputs. These are useful design hypotheses, but the broader claim that existing literature lacks such systems cannot be treated as established without a systematic, up-to-date comparison. The present thesis therefore expresses the gap conservatively as a need to test the combined workflow for the selected water classes, low-cost sensor configuration and operating context.

Prior research includes water quality indices, low-cost IoT monitoring, multivariate prediction and sensor-array pattern recognition [2]–[5]. Yet these approaches answer different questions and are not interchangeable. A water quality index aggregates selected dimensions; a threshold alert indicates a reading outside a specified range; an electronic tongue distinguishes samples using a particular sensor suite; and a classifier predicts a label learned from the dataset. The claimed contribution must be judged

against these task definitions rather than against an artificially weak “traditional monitoring” baseline.

The defensible research gap is therefore an evaluation gap: the source proposal does not supply evidence showing whether five proposed water classes can be reliably distinguished using the specified affordable sensors when whole samples, sampling days and changing acquisition conditions are held out. The protocol in Chapters 3–6 is designed to produce precisely that evidence and to report both classification performance and failure cases.

1.5 Aim and Objectives

The aim, adopted from the approved proposal, is to design, develop and validate an intelligent water identification system that uses multi-parameter sensor readings and supervised machine learning to classify selected water types. Completion is contingent on instrument calibration, a defensible reference labelling method and independent experimental validation.

The specific objectives are to assemble and verify a pH/TDS/turbidity/conductivity/temperature acquisition workflow; collect repeated readings from independently identified samples; create a documented dataset; compare suitable baseline and non-linear classifiers; deploy a consistent preprocessing-plus-inference pipeline; visualise class predictions with timestamps and data-quality messages; and assess generalisation on samples unseen during fitting or model selection. The original proposal additionally specifies target accuracy of at least 90%, classification-display latency below two seconds and affordable prototype hardware; these are acceptance targets, not observed achievements.

1.6 Research Questions or Hypotheses

RQ1: To what extent can a multi-parameter low-cost sensor array distinguish the proposed water-type classes when evaluation data come from independent physical samples rather than repeated readings of training samples? RQ2: Which of the proposed machine-learning algorithms provides the strongest balance of macro-averaged class performance, inference speed and deployment simplicity? RQ3: How

robust is the chosen model to normal operating variation, including temperature changes, sensor drift and borderline examples?

RQ4: Can a reproducible acquisition-to-display pipeline return informative classifications and actionable quality flags within the latency goal defined in the proposal? The null hypothesis for model comparison is that the candidate model does not materially improve macro-F1 over a documented majority-class or simple logistic-regression baseline on independent test samples. A meaningful statistical claim requires an explicit test design, adequate sample size and uncertainty interval.

1.7 Scope Limitations and Assumptions

In scope are sensor calibration and measurement, water-type data labelling, supervised classifier development, software integration, usability of the prediction display and controlled validation. Outside scope are comprehensive chemical or microbiological safety certification, treatment efficacy validation, regulatory approval and reliable identification of every possible contaminated water source. No water should be declared safe to drink solely on the output of this component.

The design assumes that sensors can be calibrated within their specified measuring ranges, samples are collected using a repeatable handling protocol, reference labels are assigned independently of predictor features where practical, and enough distinct physical samples can be obtained in every class. These assumptions must be examined experimentally. The thesis does not infer a product's actual performance from model descriptions or from the reference figures in the original proposal.

1.8 Individual Contribution

The individual work to be substantiated is the design and validation of a distinct water-type identification pipeline: feature selection, data protocol, training and selection of machine-learning models, model packaging, inference interface and empirical evaluation. Shared infrastructure, sensor assembly or common UI work should be acknowledged as collaborative unless authorship evidence supports a narrower attribution. For final submission, Appendix A contains an ownership and evidence log in which the candidate must identify implemented code, commits, datasets, calibration records and component-level testing actually produced.

The absence of such artefacts from the provided upload means this draft can articulate the intended individual contribution but cannot responsibly assert that every described stage has been completed. The distinction also protects the research from double counting group deliverables.

1.9 Thesis Structure

Chapter 2 reviews measurement and machine-learning literature. Chapter 3 defines the research methodology; Chapters 4 and 5 describe system design, implementation contracts and testing. Chapter 6 distinguishes available evidence from unverified results. Chapters 7 and 8 address limitations and commercial feasibility, while Chapter 9 concludes with future work. References, glossary and appendices follow.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review Method and Scope

This review is a targeted narrative synthesis rather than a completed systematic review. Sources were selected to cover measurement standards, WHO drinking-water guidance, water-quality indices, sensor-array identification, common supervised classification methods and experimental practice. Source statements originating in the proposal were retained where bibliographic identity is established; potentially incomplete or inconsistent proposal entries are not silently treated as verified studies. The candidate should extend the review with a documented database search, inclusion criteria and reference checks before final submission.

Remote-sensing reviews provide useful context for monitoring water bodies at scale, but their optical and spatial measurements are not interchangeable with the proposed in-situ sensor array [12]. The scope is purposely aligned with the research question: the central unit of comparison is water-type identification from physicochemical measurements. Studies on remote sensing and water quality indices are included as context but are not presented as direct performance baselines unless they solve a comparable prediction task. This avoids drawing misleading conclusions from differences in labels, sampling conditions, sensor modalities and validation splits.

2.2 Domain and Theoretical Foundations

pH describes the acidic or alkaline character of a solution, while conductivity measures its ability to carry electrical current. TDS readings from low-cost meters are often derived from conductivity using a conversion factor; consequently, EC and TDS may be highly correlated and should not automatically be assumed to provide independent information. Turbidity indicates optical scattering by suspended matter rather than a direct measurement of microorganisms or dissolved hazards. Temperature affects sensor responses and should be included in calibration or compensation where appropriate [1], [3].

Physicochemical measurements are imperfect proxies for water origin or processing history. Mineral water and tap water may occupy overlapping feature regions, distilled water can acquire ions through contamination or container contact, and alkaline water

can be prepared using different mineral compositions. Accordingly, class boundaries have both measurement and label uncertainty. Repeated observations from one beaker do not produce statistically independent examples simply because they appear as multiple rows in a CSV file.

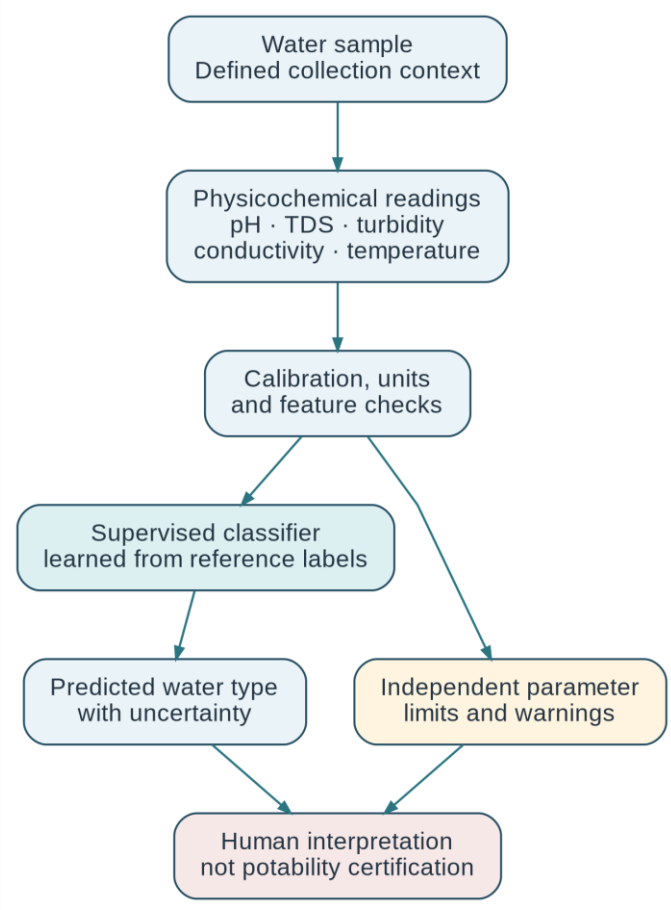


Figure 2.1: Distinction between measured features, water-type inference and independent safety checks (conceptual synthesis of [1] and [6]).

Figure 2.1 distinguishes the sensor-derived feature representation from both the categorical classifier and separate parameter-based warnings. The architecture is an original conceptual synthesis informed by water-quality guidance [1] and supervised-learning principles [6], not a reproduced experiment or drinking-water certification workflow.

A supervised classification model estimates a mapping from an input vector x = [pH, TDS, turbidity, EC, temperature] to a label y in a defined label set. Linear models provide interpretable reference decision boundaries; nearest-neighbour methods

compare a new reading with stored examples; support vector machines seek discriminating margins; tree ensembles capture non-linear interactions; and neural networks offer flexible function approximation. Every model can overfit a small or correlated dataset unless feature transformations and hyperparameters are fitted only to allowed training observations [6]–[9].

2.3 Related Research

Tyagi and colleagues review the use of water quality indices for aggregating diverse physicochemical measurements into interpretable categories [4]. These indices are valuable for comparing environmental samples against selected parameter sets, but a WQI label should not be conflated with a learned classification of distilled, tap, mineral or alkaline water. Uddin, Nash and Olbert similarly demonstrate that different index constructions can produce substantially different assessments, underscoring the importance of task-specific definitions and uncertainty [2].

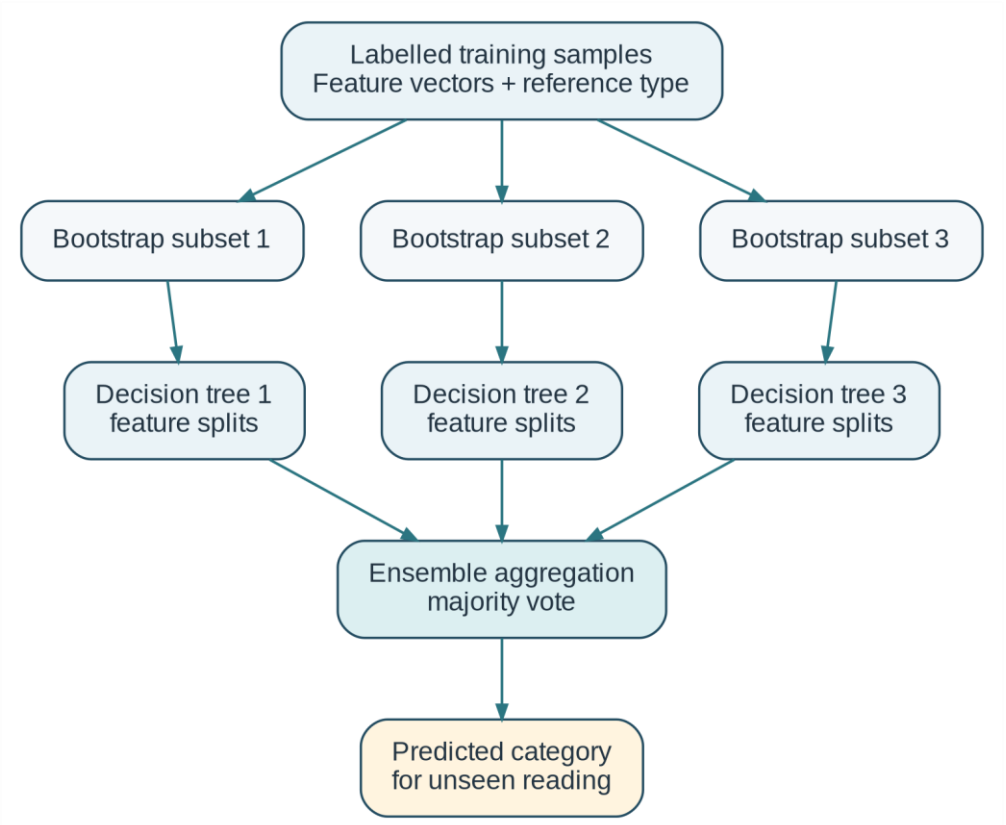


Figure 2.2: Random Forest bootstrap aggregation and voting (schematic adapted from Breiman [7]).

Figure 2.2 schematically represents the bagging principle introduced for Random Forests [7]. Each tree learns from a bootstrap sample; the ensemble combines predictions rather than choosing a single tree. The three trees are illustrative only and do not specify the final model size.

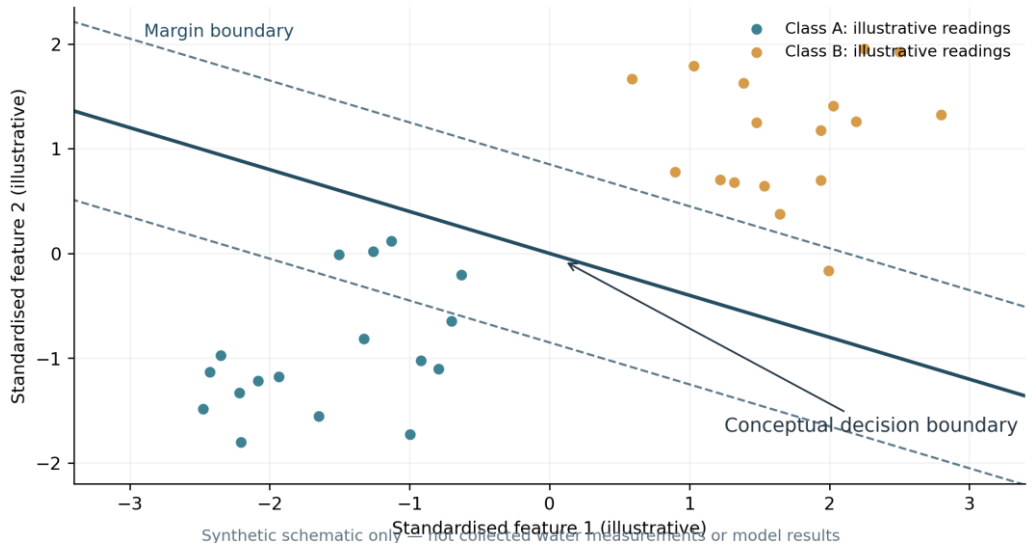


Figure 2.3: Maximum-margin classification principle of support-vector machines (schematic adapted from Cortes and Vapnik [8]).

Figure 2.3 illustrates a theoretical separating boundary with margins using synthetic points. It explains why a support-vector classifier can be a meaningful comparison model [8]; the plotted points are not project sensor data and the line is not a trained project decision surface.

The student proposal also identifies studies on IoT-based water quality monitoring, neural-network water prediction and electronic-tongue sensor systems. The electronic-tongue concept is especially relevant because it combines sensor responses to distinguish liquid samples. However, the sensor modalities, experimental conditions and reference labels may differ from an inexpensive pH/TDS/EC/turbidity array. Such work motivates multivariate recognition, not an expectation of equivalent performance without direct replication.

Pedregosa and colleagues document the scikit-learn learning ecosystem that can implement baselines, pipelines, cross-validation and metric calculation [6]. Breiman introduced random forests as a way to aggregate decorrelated decision trees [7], while Cortes and Vapnik established support vector networks [8]. These publications justify candidate models but do not prove that a particular model will dominate water-type data. Model ranking must be learned from the planned independent validation, not presumed from unrelated datasets.

2.4 Existing Systems Products and Techniques

Practical water monitoring products range from handheld TDS meters to multi-sensor telemetry devices, laboratory meters and online water management platforms. A handheld TDS device is compact and inexpensive but measures a limited proxy for ionic content. An IoT monitor can show several parameters and thresholds continuously; however, class definitions may be absent. Laboratory analysis supplies stronger traceability and hazard-specific methods at the cost of equipment, turnaround time and specialist procedures. The proposed classifier should be positioned as a complementary screening layer, not a replacement for validated analytical methods [1], [3].

Five technical approaches should be compared during model development: a majority-class baseline; a rule-based diagnostic profile defined before testing; logistic regression or a comparable linear reference; a distance- or kernel-based method such as KNN/SVM; and a non-linear ensemble such as random forest or gradient boosting. The comparison must control the data split and preprocessing steps. An apparent improvement caused by fitting StandardScaler on all observations or testing adjacent readings from the same sample would not be reliable evidence of a better model.

Table 2.1: Critical comparison of approaches

Approach	Primary output	Strength	Limitation for this study
Laboratory methods [3]	Analyte-specific measured quantities	Traceability and analytical rigor	Cost and response time
WQI assessment [2], [4]	Composite quality index	Consolidated communication	Different question from origin/type
IoT threshold monitor	Readings and warnings	Continuous acquisition	Limited categorical insight
Electronic-tongue pattern recognition	Liquid-class prediction	Multivariate classification concept	Different hardware and labels
Proposed system	Predicted type + quality flags	End-to-end low-cost workflow	Needs independent validation

2.5 Gap Synthesis and Conceptual Framework

The conceptual framework has four linked elements: measurement quality determines feature reliability; label quality and sample diversity determine what the model can learn; validation design determines whether reported performance reflects real generalisation; and interface design determines whether predictions are interpreted safely. A failure in an upstream element cannot be fixed merely by displaying a confident downstream classification. In particular, a model can obtain a high score on correlated rows while remaining unreliable on a new physical sample.

Figure 3.1 operationalises this framework. The proposed contribution is the integrated evaluation of sensor acquisition, preprocessing, independent-sample classification and result reporting. Its novelty should be characterised as a context-specific, reproducible implementation and validation of known techniques, rather than as invention of a new machine-learning algorithm. Novelty will only be demonstrated by evidence of improved workflow, meaningful performance or useful deployment characteristics under specified conditions.

3 RESEARCH METHODOLOGY

3.1 Research Design

The proposed methodology combines design science and controlled experimental evaluation. A research artefact is defined as the integrated pipeline comprising sensor acquisition, labelled water dataset, preprocessing, trained predictor and reporting interface. Design work establishes hardware and software contracts; experimental work tests class separability, generalisation and technical performance. The method follows the iterative phases in the student's proposal: hardware and firmware, data collection and preprocessing, model development, system integration and end-to-end validation.

Two forms of validation are kept distinct. Software verification asks whether inputs, units, packet formats, preprocessing and output semantics are implemented according to specification. Research validation asks whether the learned model makes useful predictions for new physical samples under the actual sampling regime. Passing unit tests cannot demonstrate scientific accuracy; obtaining a good model score cannot demonstrate robust packet parsing or safe failure behaviour. This distinction is particularly important for an embedded sensing project.

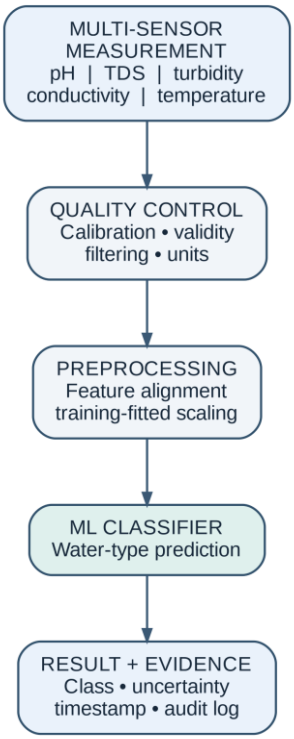


Figure 3.1: Conceptual water-type identification pipeline

3.2 Data Participants or Study Materials

The proposed study materials are representative samples of distilled, tap, alkaline, mineral and deliberately prepared, non-hazardous “contaminated” or altered water, subject to an approved sampling and labelling protocol. The final list of classes must reflect what was actually collected and must be supported by a documented definition for every label. The protocol should include multiple independent containers or source batches in each class, distinct sampling days and realistic environmental variation. Physical samples, not time-series readings, are the primary units of independence.

The planned hardware comprises an ESP32-class controller with pH, TDS, turbidity, electrical conductivity and temperature sensors as defined in the proposal. The proposal’s financial table lists pH, TDS and turbidity modules but does not separately cost conductivity and temperature sensors. The final bill of materials therefore requires reconciliation with actual purchases and installed parts. Device model, calibration standards, probe age, sensing interval, analog reference and signal conditioning should be documented, because each can influence measurement uncertainty.

Each record should carry a sample_id, collection_time, source_batch, declared_type, label_method, calibration_id, device_id, pH, TDS_ppm, turbidity_NTU, EC_uScm, temperature_C and quality_flags. Personally identifying household information is unnecessary and should be avoided. Where the same sample generates many repeated measurements, those measurements should remain grouped during statistical splitting.

Table 3.1: Planned acquisition dataset schema

Field	**Type / units**	**Reason**
sample_id	Pseudonymous identifier	Group independent observations
source_batch	Categorical	Track batch/collection dependence
declared_type	Class label	Supervised training target
label_method	Controlled vocabulary	Reference-label provenance
pH	Numeric (pH scale)	Acidity / alkalinity
TDS_ppm	Numeric (ppm)	Dissolved-solids proxy
turbidity_NTU	Numeric (NTU)	Optical scattering proxy
EC_uScm	Numeric (µS/cm)	Electrical conductivity
temperature_C	Numeric (°C)	Response compensation
calibration_id	Identifier	Traceability
quality_flags	List / code	Missing, unstable, out-of-range data

3.3 Experimental or Study Procedure

Before sampling, the candidate should establish and document calibration procedures using manufacturer-appropriate standards and reference instruments where available. Standard pH buffer checks, conductivity solutions and appropriate turbidity standards should be logged with time and operator. Stable acquisition requires probe rinsing, defined immersion depth, settling time, sample temperature and a fixed acquisition interval. In the proposal a one-second example interval and a two-to-three-minute acquisition window are suggested; these are planning assumptions, not evidence of the implemented settings.

For each labelled physical sample, repeated readings are acquired, sanity-checked and retained with provenance. Measurements should not be indiscriminately averaged before reviewing drift, spikes or initial stabilisation. An explicit rule, established before model selection, determines whether the central tendency, dispersion or full temporal sequence is used for classification. Duplicate removal and outlier filtering must be documented; the latter must not silently remove legitimate difficult cases that the deployed system will later encounter.

The dataset is partitioned using groups that prevent readings from the same water sample or collection batch appearing in both training and test sets. A training subset supports model fitting and inner cross-validation; a held-out subset is reserved until the entire model and preprocessing pipeline have been selected. If data allow, a temporal or new-device holdout offers stronger evidence about future deployment. Random row-level splitting would inflate the apparent sample size and can produce optimistic results.

All candidate models should receive exactly the same feature set and evaluation split unless a comparison explicitly studies feature selection. The experiment records class balance, missingness, calibration changes, algorithm version, hyperparameter grid, random seed and per-sample predicted probabilities or scores. After selection, the final pipeline is packaged and subjected to repeatable inference tests with valid, boundary, missing and out-of-range packets.

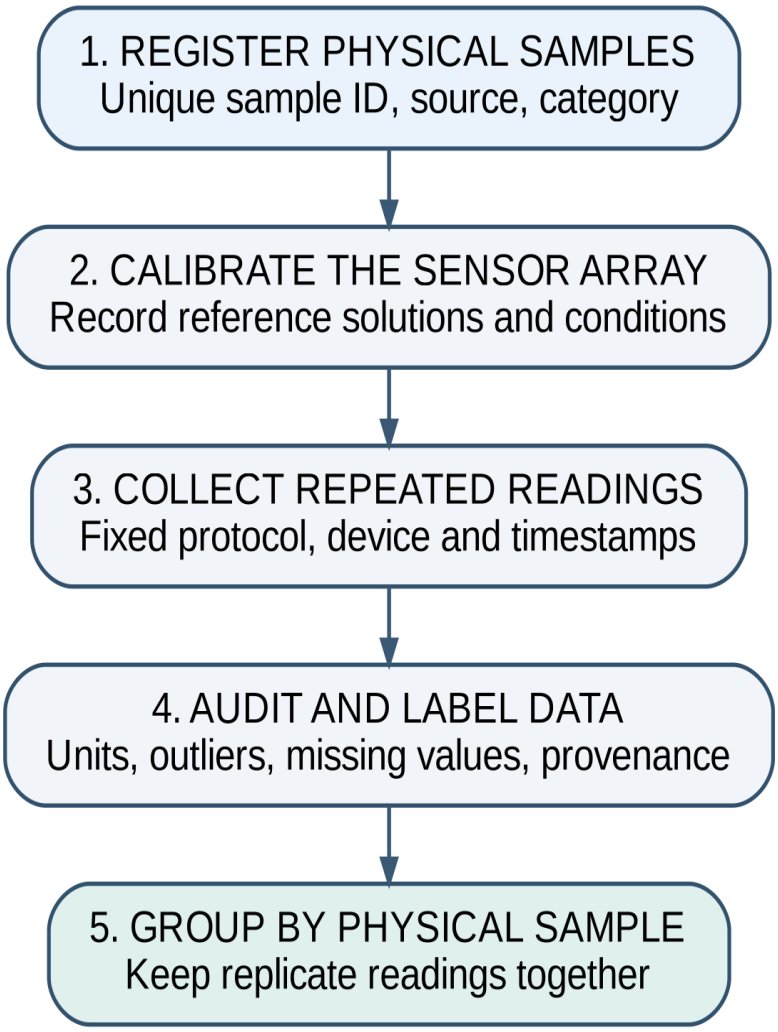


Figure 3.2: Proposed sample and calibration protocol

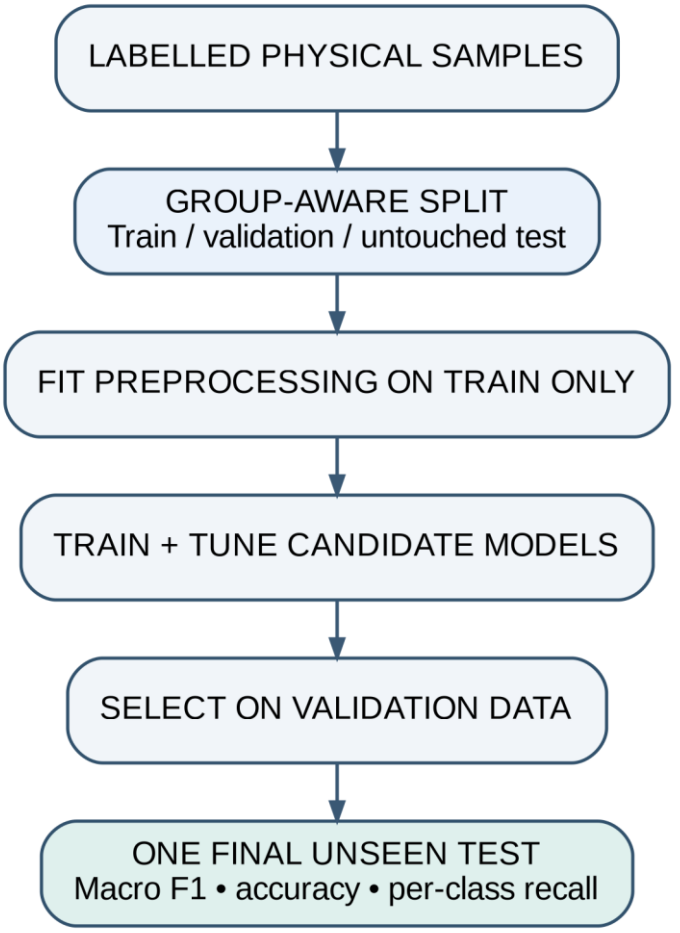


Figure 3.3: Leakage-resistant validation flow

3.4 Measures and Evaluation Metrics

The primary research metric should be macro-averaged F1 because it gives equal weight to each class regardless of frequency. Accuracy, balanced accuracy, per-class precision, recall and confusion matrices provide complementary interpretation. For a class c, precision = TPc/(TPc + FPc), recall = TPc/(TPc + FNc) and F1c = 2·precision·recall/(precision + recall). Macro-F1 is the mean of the class-wise F1 values. Zero-denominator conventions should be stated in the experiment script.

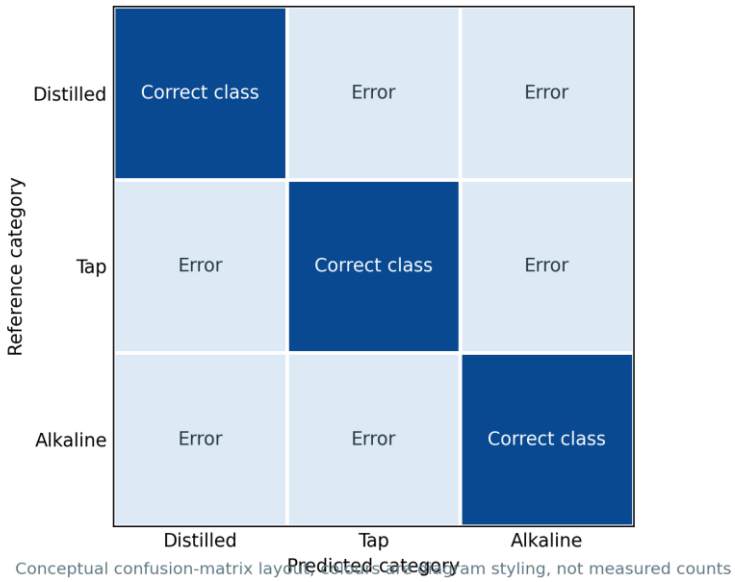


Figure 3.4: Interpretation of a multiclass confusion matrix (conceptual layout based on evaluation conventions in [6]).

Figure 3.4 explains the relationship between reference labels and predicted classes. Diagonal entries represent correct assignments and off-diagonal entries represent class confusion. The cell shading is instructional, not experimental evidence; final results require actual sample counts and class-specific recall [6].

Operational measures include end-to-end latency from acceptance of a complete sensor packet to display of the prediction, packet-loss or invalid-message rate, repeatability of measurements, memory or CPU usage, and successful recovery after transport interruption. Latency should be summarised using median and 95th percentile rather than just a single run. The proposal sets targets of at least 90% accuracy, under two seconds per display update and a hardware budget below LKR 25,000; the source document also lists an indicative purchase table of LKR 13,500. A threshold is not an observed result.

Model uncertainty needs attention. A predicted probability is not necessarily calibrated, particularly for small datasets or out-of-distribution samples. Any confidence threshold or abstention rule should be selected using training-side validation, then assessed unchanged on the held-out set. Coverage (fraction of samples receiving a class) and selective error rate can make the cost of abstention explicit.

Safety-language tests should verify that a water-type label is never presented as a drinking-water approval.

Table 3.2: Metrics and acceptance criteria

Measure	Calculation / procedure	Status
Accuracy	Correct predictions / all independent held-out samples	Proposal target ≥ 90%; unverified
Macro-F1	Mean F1 across reported classes	Primary evaluation metric; pending
Per-class recall	TP / (TP + FN) per label	Report with uncertainty
Latency	Packet accepted to display; p50 and p95	Proposal target < 2 s; pending
Robustness	Degradation under planned drift / noise cases	Pending controlled experiment
Coverage	Classified samples / all valid evaluation samples	Pending abstention evaluation

3.5 Data Analysis

Descriptive analysis will report the number of independent physical samples per class, repeated measurements per sample, distributions and ranges for every feature, missing-value rates and cross-feature correlation. Because TDS and EC may reflect the same physical signal, correlation diagnostics and feature ablation can determine whether including both adds information. Group-aware descriptive summaries should prevent a few heavily sampled containers from dominating the aggregate statistics.

Within training data, preprocessing should be implemented as a single version-controlled pipeline. Where scaling is necessary, transformation parameters are fitted only on the training fold and then applied to the corresponding validation fold. Hyperparameters may be tuned using grouped cross-validation or an equivalent nested design if sample size permits. A single untouched test set is evaluated once after the candidate and decision threshold have been finalised. The analysis report should provide confidence intervals computed by resampling at sample_id or source-batch level, not by assuming all repeated sensor rows are independent.

Comparative results should prioritise effect sizes and uncertainty over the claim that a small numerical difference is “significant”. Error analysis should review confusion pairs, unstable sensor conditions, overlapping feature vectors and systematically difficult source batches. If a class has too few independent samples to support inference, the limitation must be reported rather than hidden in an overall accuracy number.

3.6 Ethics Privacy Security and Responsible AI

Most planned inputs are water properties rather than personal data. Nonetheless, source addresses, household identifiers, precise private locations and usernames can become personal information; they should be excluded or pseudonymised. Stored telemetry should use defined access permissions, retention periods and encrypted transport where applicable. External network connectivity introduces risks of altered sensor packets, replayed data and unauthorised model updates, so messages should be validated and deployment credentials managed outside code repositories.

False reassurance is the primary ethical hazard. The absence of detectable differences in the five chosen parameters does not establish microbial safety or absence of dissolved toxic chemicals. The dashboard should state “predicted water type” and show sensor-based limitations; it should avoid unconditional “safe to drink” labels. On missing calibration, implausible measurements or low confidence, the preferred behaviour is to abstain and recommend independent testing as appropriate [1].

Water collection and deliberately altered test samples must follow local laboratory and environmental procedures. “Contaminated” water should not require creating toxic mixtures; safe simulated perturbations are preferable where academically valid. Any release of sensor datasets should remove sensitive location information and document label uncertainty. The human researcher remains accountable for decisions and for disclosure of AI-assisted writing or coding.

3.7 Reproducibility and Research Artefacts

A reproducible project folder should contain the raw sensor packets; a data dictionary; calibration logs; sampling protocol; anonymised sample-to-batch mapping; cleaning scripts; a data split manifest; fitted-model configuration; training code; test scripts; and a versioned environment description. Each derived table or figure should be traceable to a command and source dataset, with a recorded date and software version.

To support verification of the individual contribution, hashes or repository commit identifiers should link results to immutable input files. Record the exact definitions of each water class and all post-collection relabelling decisions. The final report should identify which artefacts are owned by the student and which are shared with the group. Appendix A provides a suggested evidence register for this purpose; until real artefacts are supplied its fields are intentionally unfilled.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The proposal defines five core functional capabilities: acquiring pH, TDS, turbidity, conductivity and temperature measurements; correcting or filtering sensor values; classifying a sample into a predefined water-type label; presenting measurements, class and timestamp; and logging the history for review. The proposed system must also reject malformed data, preserve units, expose stale measurements and identify calibration status. These elaborations translate research and safety requirements into testable engineering behaviours.

Non-functional requirements include classification performance, latency, robustness to normal noise, usability, affordability and modularity. The proposal specifies at least 90% accuracy, a response below two seconds and a hardware ceiling below LKR 25,000. Additional requirements for traceability, class-label versioning and safe failure are necessary because a water monitoring system may affect real decisions. The table below distinguishes proposal targets from requirements introduced here as design safeguards.

Table 4.1: Functional and non-functional requirements

ID	Requirement	Evidence / test
FR-01	Read five measured parameters with units	Sensor packet contract
FR-02	Validate calibration, missing and range conditions	Boundary and calibration tests
FR-03	Predict only supported labels	Model output schema and tests
FR-04	Display readings, class, time and caveat	UI acceptance inspection
FR-05	Persist traceable prediction logs	Log reconciliation test
NFR-01	Accuracy target ≥90% (proposal)	Independent-sample test
NFR-02	Latency target <2 s (proposal)	p50/p95 measurement
NFR-03	Hardware budget <LKR 25,000	Invoice-based bill of materials
NFR-04	Avoid unsafe reassurance	Abstention and UI copy review

4.2 Component Architecture within the Complete Solution

The intended architecture consists of six logically separate functions: physical sensing, microcontroller acquisition, message transport, ingestion and validation, machine-learning inference and result presentation. Every transfer should have an explicit schema, unit convention and error response. A sample identifier and timestamp must survive the whole chain so that displayed results can be matched to source measurements. The classifier is an independent service boundary that can later support other group components through documented interfaces.

The proposed architecture should not be interpreted as proof that these blocks have been implemented or deployed. In particular, the original proposal allows either serial or Wi-Fi transport and suggests either a desktop or web dashboard. The actual choices must be updated from source code and deployment screenshots once available. Figure 4.1 shows a technology-neutral logical decomposition that is compatible with those proposal alternatives.

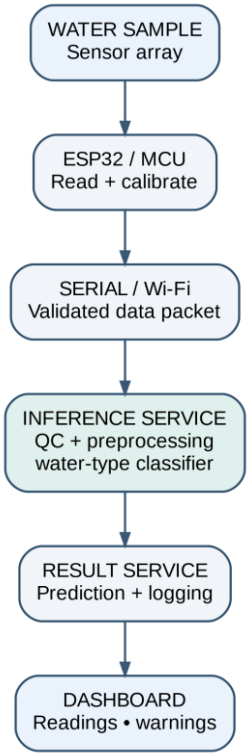


Figure 4.1: Proposed logical architecture and interfaces

4.3 Design Alternatives and Decisions

An ESP32-class microcontroller offers an accessible combination of analog inputs, processing capability and optional Wi-Fi, while Arduino alternatives could supply simpler wired acquisition. The proposal favours ESP32, but the final choice should depend on sensor-voltage compatibility, ADC behaviour, transport requirements and available hardware. A host-computer ML service reduces the need to compress the classifier onto the microcontroller and enables transparent retraining. Edge inference could be explored if network dependency or latency becomes unacceptable.

For models, logistic regression provides a transparent baseline, KNN can model local similarity but depends on scaling and inference set size, SVM can offer non-linear decision boundaries with tuning, and random forest/gradient boosting can capture interactions with relatively little feature engineering. A neural network should only be pursued if there is enough independent training data to justify its capacity. The final algorithm must be chosen using the same split and metrics, not because an algorithm is fashionable or named in the proposal.

The UI decision should preserve clarity about prediction uncertainty and input problems. Gauges or traffic-light colours without text risk implying a safety decision that the model has not validated. A compact class label, measurement table, timestamp, status flag and a caution on scope are better aligned with the study. Planned interfaces should support an “unclassified” state rather than forcing a potentially invalid sample into a known category.

4.4 Detailed Design

An acquisition packet can be expressed as a versioned JSON object with fields for schema_version, device_id, sample_id, measured_at, pH, tds_ppm, turbidity_ntu, conductivity_us_cm, temperature_c and calibration_id. The backend validates type, range, units, timestamp format and the presence of all required features. It should not silently replace null values with zero, because such an imputation can produce false water-type signatures.

A complete inference result contains the original sample_id, accepted_at, predicted_class or null, model_version, confidence if meaningful, quality_flags and latency_ms. The accepted class-label vocabulary is maintained in one versioned

mapping. This preserves consistency across data collection, model training, API responses, UI labels and evaluation plots. Error responses distinguish invalid measurements, model unavailable, unrecognised class labels and transport timeouts, with each failure logged as an event.

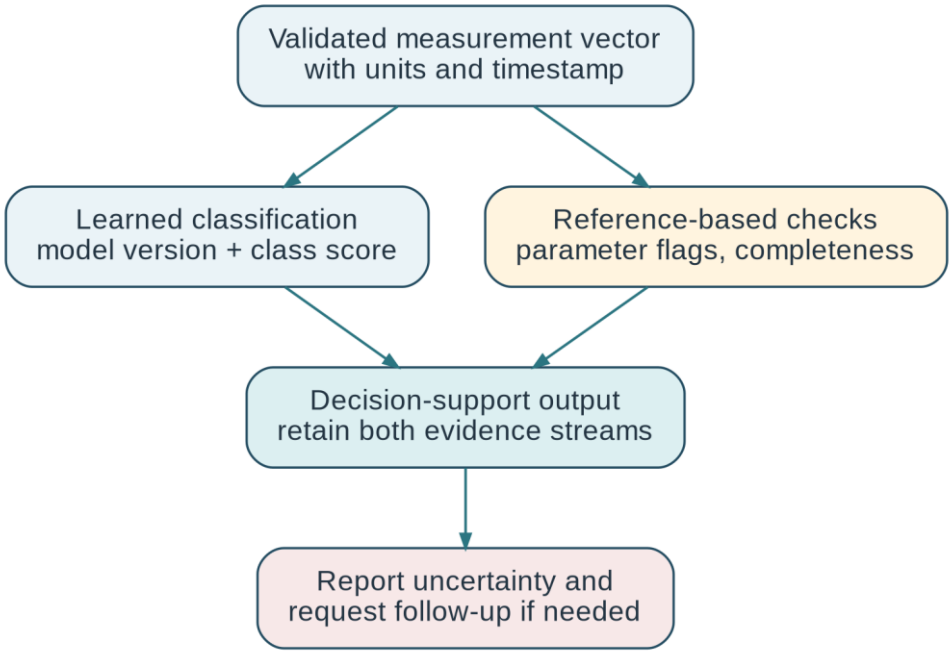


Figure 4.2: Parallel learned classification and parameter-rule interpretation (conceptual synthesis of [1], [6], and [13]).

Figure 4.2 keeps learned predictions separate from deterministic flags and validation status. Preserving both branches supports traceability and protects against the false inference that a favourable type label alone demonstrates potability [1], [6], [13].

The proposed prediction path should apply the identical fitted preprocessing objects used during training. A versioned model bundle stores expected feature ordering, transformation parameters, class labels and metadata. The server must check schema_version and expected columns before invoking the classifier. If a feature is absent, stale or out of sensor range, the response should withhold a definitive class unless a validated missing-feature strategy exists.

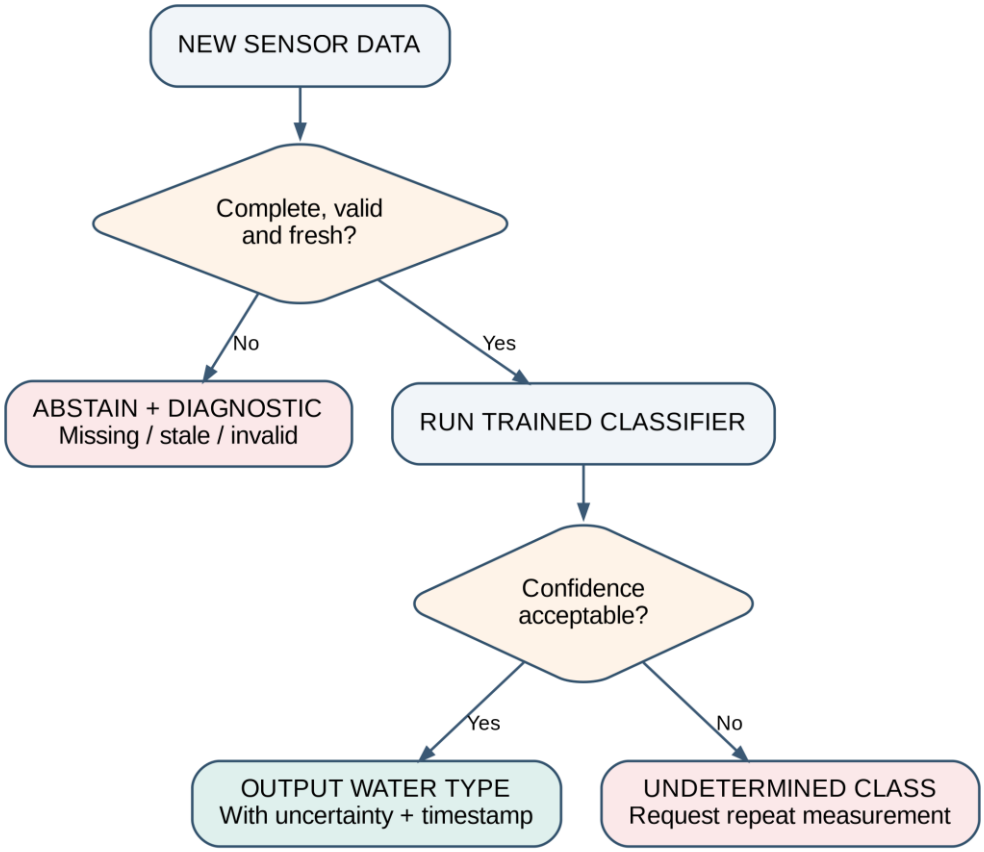


Figure 4.3: Safety-oriented decision and abstention pathway

4.5 Standards and Professional Practice

Water measurement procedure should be informed by established analytical methods rather than unverified consumer specifications [1], [3]. Software design should follow traceable requirements, separate training and deployment configurations, and retain versioned tests. Dataset documentation should include class provenance, physical sample grouping, calibration records and limits of generalisation. A security review should cover transport, credentials and audit logs if the system is network-connected.

The intended use must remain explicit: this is a research classifier, not a certified drinking-water safety instrument. Any automated purification actuator would require independently verified control thresholds, fail-safe states and treatment safeguards. ISO/IEC 25010:2023 supplies a recognised product-quality framework [14].

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development Environment and Technology Stack

The original proposal describes firmware authored through the Arduino IDE for an ESP32-class board, CSV logging, preprocessing with Python and Pandas, classifiers implemented using scikit-learn and potentially TensorFlow/Keras, plus a desktop Tkinter or web Streamlit dashboard. This collection is the planned technology stack. No executable repository, firmware binary, trained model, screenshots or measured integration logs were included with the supplied material, so deployed versions and final package choices cannot yet be verified.

A suitable reproducible environment would pin Python and library versions, use a virtual environment or lockfile, distinguish raw data from derived model artefacts and provide one command for training and another for inference. The codebase should separate firmware, data ingestion, feature transformation, model evaluation, inference, interface and tests. Model serialization with joblib or another compatible format requires trusted artefact storage, because loading untrusted pickle-based files is unsafe.

Device-specific configuration should be stored outside the model and source code. Sensor coefficients, serial port names, baud rate, Wi-Fi endpoints and sampling intervals vary between deployments, whereas class schema and validation rules should remain versioned. A deployment manifest should record the hardware revision, sensor model, firmware revision, backend commit and model-training dataset revision. These interface, configuration and data-dependency controls respond to the maintenance risks described in deployed ML systems [13].

5.2 Component Implementation

Firmware implementation should initialise each sensor, record raw ADC values where appropriate, apply calibration coefficients and emit a complete observation packet. The controller should separate an acquisition failure from a legitimate zero or low reading. An unstable probe or ADC saturation should produce a status flag, not a plausible-looking fabricated value. The planned unit tests can exercise packet generation using recorded sensor fixtures even without a connected board.

Data processing begins by checking units and schema, identifying duplicates, linking repeated observations to physical sample IDs and documenting any exclusions. Feature transformation and classification are then composed into a fitted pipeline. A script should produce a frozen, reproducible split manifest before hyperparameter tuning. The selected model is exported only after its configuration and data dependencies are recorded; later training should produce a new version rather than silently replacing an operational model.

An inference service accepts a validated numeric vector and returns an enumerated water label or an abstention response. UI rendering occurs only after a consistent response has been obtained. The interface should display the most recent successful reading while visibly indicating stale or disconnected status; showing an old prediction as if it were current would undermine the real-time requirement. All logs should include enough fields to reconstruct the cause of any unusual result.

5.3 Integration with the Other Components

Integration should be specified at the boundary rather than assuming another component consumes unspecified fields. The identifier can expose predicted_type, class_set_version, recorded_parameters, quality_flags, measurement_time and model_version. A water-treatment adviser or safety-report component can then decide whether to request further verification. No external component should infer that a predicted “distilled” or “tap” label means microbiological or chemical safety.

The group-level architecture described in the proposal includes automated, context-aware action as a possible future feature. This creates a need for a deliberately limited contract: the classification service reports observed values and evidence status; a distinct rules engine, verified against appropriate standards, makes any treatment recommendation. An actuator should not be triggered solely by a potentially uncertain ML label without an approved safety interlock.

For integration testing, each message type should have a contract fixture: normal readings, invalid units, unsupported class, missing calibration identifier, stale timestamp, transport disconnect and model-version mismatch. Test harnesses can replay these packets into the backend while capturing the rendered UI response and logged event. Group members should agree on ownership of cross-component failures.

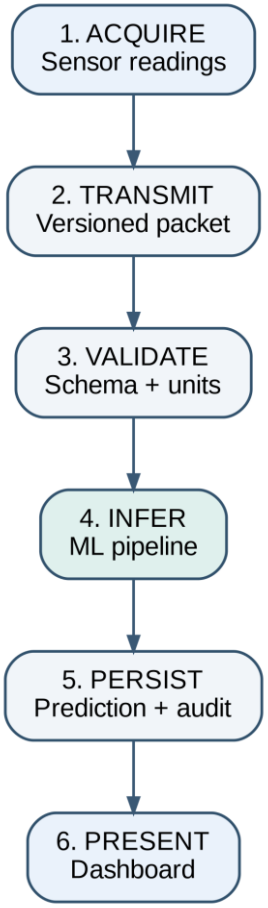


Figure 5.1: Proposed end-to-end integration sequence

5.4 Verification and Software Testing

A robust test hierarchy starts with sensor-conversion unit tests, feature-ordering tests and model-bundle compatibility tests. It then tests serial or network ingestion, schema validation, dashboard rendering and persistence. Research-level tests are separate: sample-aware cross-validation, independent test classification, noise or drift stress cases and confidence/abstention evaluation. This distinction permits a meaningful trace from requirement to evidence and avoids overstating software integration as experimental validation.

Test cases require preconditions, input values, expected output, pass/fail criteria, observed outcome, execution date and evidence reference. The supplied proposal describes objectives and target thresholds but does not include executed test records. Accordingly, Table 5.1 is a proposed test matrix rather than a report of passed tests.

The student should replace the blank results fields with logs or screenshots linked in Appendix A.

Table 5.1: Proposed verification test matrix

Test	Input / scenario	Expected result	Execution
TC-01	Valid five-parameter sensor packet	Prediction or justified abstention; timestamp	Not supplied
TC-02	Missing pH field	Structured validation error	Not supplied
TC-03	Out-of-range sensor value	Rejected or flagged, never silently clipped	Not supplied
TC-04	Training/inference feature-order mismatch	Fail closed with schema error	Not supplied
TC-05	Reconnect after network loss	Visible stale state and recovery log	Not supplied
TC-06	New independent physical sample	Recorded class + confusion record	Not supplied
TC-07	Low confidence / unknown mixture	Abstain according to fixed rule	Not supplied
TC-08	50 consecutive packets	Latency distribution and dropped count	Not supplied

5.5 Deployment and Operation

A deployable package should include clear installation steps, supported sensor ranges, calibration procedure, data collection settings, environment dependencies, model checksum, start-up health checks and a recovery process. Monitoring should capture recent sensor faults, inference failures, model version, packet throughput and latency. Operational users need a visible indication when the most recent measurement is too old to represent current water conditions.

Because water samples vary over time and between devices, operational maintenance requires periodic calibration, drift checks and an agreed retraining trigger. A retrained model should be evaluated on an untouched reference collection and promoted only when objective acceptance criteria are met. Rollback must remain available. At this draft stage these constitute operational requirements, not proof of a production deployment.

6 RESULTS AND VALIDATION

6.1 Research Results

The provided material is a research proposal rather than an experimental log or final results report. It includes objectives, planned algorithms, measurement parameters, a schematic methodology and numeric entries in comparative appendices, but it does not identify raw datasets, test sample counts, trained model artefacts, executable evaluation notebooks or a reproducible calculation method for those entries. Therefore, none of the proposal's comparison values can be represented here as independently validated classifier accuracy, precision, recall or F1.

The results section is intentionally evidence-led. Once experiments are complete, it should begin with the number of independent physical samples by class; sampling dates and batches; missing-data and calibration exclusions; exact training/validation/test group counts; selected model and hyperparameters; confusion matrix; per-class scores; macro-F1; accuracy with confidence intervals; inference latency; and a description of any abstention cases. Supporting scripts and independent test predictions should be available to the supervisor.

Until these materials are supplied, the correct research finding is that quantitative model performance is **not established by the uploaded documents**. Reporting 90% accuracy as an achievement would misstate a proposed requirement. Likewise, assigning performance scores to individual algorithms from illustrative appendix tables would risk making unsupported claims of completed experiments.

Table 6.1: Evidence availability register

Evidence item	Available in uploaded sources?	Permissible thesis statement
Research proposal and objectives	Yes	Planned aim and scope established
Conceptual workflow and architecture	Yes	Proposed design established
Raw labelled dataset and split IDs	No	No dataset statistics yet
Training scripts and fitted models	No	Implementation not verified
Confusion matrix / held-out predictions	No	Accuracy/F1 not established
Firmware and device calibration logs	No	Hardware performance unverified
Integration test traces and UI evidence	No	Deployment and latency not established
User acceptance or stakeholder study	No	Usability outcomes not established

6.2 Product and Technical Results

The proposal supplies a planned bill of materials and defines the desired end-to-end workflow, but lacks assembly photographs, pin-level wiring records, firmware commits, device IDs, sensor calibration traces and API screenshots. As a result, the present document cannot certify the completeness of the physical prototype. Once available, evidence should include a photograph with component annotations, a circuit or wiring diagram, a sample serial output, a calibrated measurement log and a deployed dashboard view showing a real timestamp.

For technical reporting, latency should be measured at clearly defined points: packet receipt, validation completion, inference completion and UI update. A single stated latency target is not a measurement. A reproducible benchmark should disclose the host device, connection mode, packet size, model type, trial count and distribution of timings. The same applies to current draw, memory consumption and error recovery: proposed attributes do not constitute observed performance.

6.3 User or Stakeholder Validation

No conducted usability interviews, survey responses or stakeholder evaluation records are included in the supplied files. The final validation can employ short task-based observations in which intended users identify the predicted class, read the underlying measurements, recognise an out-of-range warning and explain whether the interface claims drinking-water safety. A suitable protocol should record participant recruitment, consent, tasks, success criteria, time on task and anonymised qualitative feedback.

Usability findings should differentiate preference from correct interpretation. A participant may find the dashboard attractive while misunderstanding the difference between a water-type prediction and certified water safety. The latter should be an explicit comprehension check. The study should report any changes made to labels, warnings or visual hierarchy in response to genuine user evidence.

6.4 Integration Results

Integration performance has not been demonstrated by the uploaded documents. The final section should report tested interface versions, device-to-backend connectivity, message validation outcomes, successful inference counts, loss/reconnection tests, log completeness and the behaviour of dependent team components. An architecture diagram alone does not prove correct communication or performance. The acceptance record should identify exactly which cross-component features were completed and which remain prototype-only.

Any claim about automatic purification, mineral dosing or water-safety advice would require separate evidence belonging to those subsystems, with documented source and student ownership. In the absence of such evidence, this individual thesis restricts results discussion to the planned water identifier and the explicit contracts by which other systems might use its output.

6.5 Summary of Findings

The documentary findings currently supported are that a clearly scoped water-type classification problem has been proposed; five sensor parameters and multiple candidate ML algorithms were specified; a five-phase methodology and logical

architecture were drafted; and measurable accuracy, latency and cost targets were stated. The documents do **not** establish achieved accuracy, a winning algorithm, a complete deployed classifier or successful field validation. These are material gaps, not editorial omissions.

A submission-ready results chapter requires the author to attach and verify experiment evidence, update the abstract with principal findings, compare achieved metrics to targets and re-evaluate the conclusion. The validation matrices in Appendix A are prepared so that the missing evidence can be inserted without redesigning the chapter.

7 DISCUSSION

7.1 Interpretation of Findings

The main analytical implication of the proposed architecture is that identification quality is inseparable from measurement and labelling quality. A classifier trained on highly repetitive recordings may appear precise while learning idiosyncrasies of containers, calibration events or acquisition days. The recommended sample-aware split addresses this threat but may lower the score relative to a row-level random split. Such a decrease would be scientifically useful because it better approximates genuine future use.

Water classes can overlap in the selected feature space. The interpretation of any eventual confusion matrix should therefore consider whether two labels have distinguishable physicochemical signatures under the measurement resolution and calibration conditions. If not, a hierarchy of broader classes, an abstention output or additional sensor modalities may be more defensible than continuing to tune larger models. Research success should include identifying conditions where classification cannot be supported.

7.2 Comparison with Literature Baselines and Existing Solutions

The proposal's emphasis on categorical intelligence is meaningfully different from a simple threshold alert, but categorical models already exist across sensor-based liquid identification. The defensible comparison is whether the specific low-cost sensing configuration can deliver dependable, transparent classification for the proposed water labels with acceptable cost and latency. Related WQI research underscores the need to distinguish tasks rather than compare numerical scores across incompatible definitions [2], [4].

Once actual experimental outcomes are available, comparison should be reported against a majority predictor, pre-declared threshold profile and one simple statistical classifier trained on the same samples. Improvements should be supported by independent held-out performance and uncertainty, with transparent disclosures of feature set, class definitions and split. Claims that random forest outperforms SVM in general would not follow from a single water dataset.

7.3 Research Contribution

The intended research contribution is a repeatable protocol joining controlled multi-sensor measurements, traceable water-type labels, group-aware validation and operational inference tests. Its central methodological contribution is to make the hidden dependencies of a sensor-ML system explicit: calibration, class provenance, grouping, transformation reuse, uncertainty and safe interpretation. If executed, such a protocol can support more trustworthy evidence than an isolated high model score.

Novelty must be stated proportionately. The proposal does not introduce a new learning algorithm; instead, it proposes an applied synthesis of known sensing and machine-learning techniques. Its final contribution can be evaluated on context, affordability, usability and experimental design, provided these are independently demonstrated rather than claimed in advance.

7.4 Product Contribution

The planned product contribution is a modular interface that turns measured sensor readings into a water-type estimate with explicit supporting measurements, version information and quality flags. Its potential usefulness is that an operator can see more than a collection of raw numbers while retaining the ability to inspect the readings that generated the label. The system is especially appropriate as a prototype for controlled demonstrations and preliminary screening, not as a stand-alone potability instrument.

Potential connections to a broader purification project should be documented as interfaces and proposed extensions. The identifier cannot guarantee safe actuator behaviour and should not independently command chemical dosing. If integrated later, a verified control layer must define authorised actions and handle disagreement between sensors, classifiers and treatment requirements.

7.5 Limitations and Threats to Validity

Construct validity concerns whether the ground-truth labels correspond to physically meaningful water categories. The name “mineral water”, for example, may describe a commercial product rather than a stable interval of sensor values. Internal validity is threatened by sensor drift, calibration procedures, collection order, label leakage and

inconsistent cleaning. External validity is threatened by source variability, new brands, unseen contaminants, seasonal changes and device differences.

Statistical validity depends on the number of independent physical samples, not the number of timestamped rows. Related readings from the same source reduce effective sample size, and repeated tuning against a single holdout can create hidden overfitting. Small class counts make confident statements about precision or recall difficult. The final thesis should give confidence intervals and describe all exclusions, failed samples and class-specific mistakes.

The current documentary limitation is substantial: no author-supplied raw experimental, implementation or acceptance data accompany the draft. Thus a final-grade claim of validated performance would be unsupported. The recommended remedy is not to invent numbers but to obtain source artefacts and re-run the analysis reproducibly.

7.6 Implications

For researchers, this design highlights why inexpensive sensing projects need disciplined sample provenance and leakage-resistant validation. For product developers, it suggests that a class prediction should be accompanied by timestamps, sensor-status flags and model metadata. For users, it reinforces that water-type identification is not equivalent to hazard testing. For future group integration, clear boundaries between sensing, classification, recommendation and actuation are essential.

Future experiments should establish whether this sensor suite generalises to new sources and whether additional measurements or a revised label taxonomy are needed.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value Proposition and Target Users

The proposal identifies possible users in households, consumer electronics, industrial quality assurance and environmental field screening. A realistic early adopter is a researcher, student laboratory or prototype development team needing repeatable indication of selected water types under controlled conditions. Expanding to consumers or regulated industrial applications would require evidence of reliability, suitable calibration support, product safety and compliance.

The value proposition is not that the device replaces laboratories. Instead, it aims to combine several affordable measurements and provide a comprehensible category with context and logs. Any claimed saving in time or expense should be measured against a defined alternative and should include calibration, consumables, maintenance and operator effort, not only sensor purchase prices.

8.2 Market and Competitive Position

Handheld conductivity/TDS meters, multi-parameter probes and laboratory services occupy different positions in cost and analytical depth. The proposed classifier would sit between a single-parameter consumer meter and a more capable analytical system, seeking usefulness through multi-parameter patterns and clear user-facing interpretation. However, its classification labels are not necessarily comparable to certified test outputs, so marketing must avoid inappropriate safety guarantees.

A credible competitive study should collect current product prices, supported parameters, accuracy specifications, calibration requirements, warranties and local distribution information. This draft does not invent market share or competitor prices. The research proposal lists candidate segments and B2B licensing ideas but contains no validated customer interviews or willingness-to-pay data.

8.3 Business or Sustainability Model

The proposal suggests potential direct hardware sales, OEM licensing and cloud analytics. For a university prototype, the shortest path to viability may be a demonstrator kit and service package including sensor calibration instructions, model versioning and interpretation guidance. A future subscription offering would require

secure connectivity and transparent policies for collection and retention of sensor histories.

The indicative hardware budget from the proposal lists an ESP32 at LKR 2,500, pH module at 4,500, TDS module at 2,000, turbidity module at 500, wiring at 1,000, power supply at 5,000 and miscellaneous components at 2,000, totalling LKR 15,500. Since conductivity and temperature parts are discussed in the narrative but not separately listed in this budget, and actual invoices are not provided, the figure should be treated as a historical estimate, not as the final product cost.

Table 8.1: Indicative prototype cost plan reproduced from proposal

Item	Proposed cost (LKR)	Cost status
ESP32 microcontroller	2,500	Estimated
pH sensor module	4,500	Estimated
TDS sensor	2,000	Estimated
Turbidity sensor	500	Estimated
Breadboard and wires	1,000	Estimated
Power supply	5,000	Estimated
Miscellaneous	2,000	Estimated
Total	15,500	Excludes separately unpriced components

8.4 Scalability Intellectual Property and Compliance

Scalability at the technical level depends on the ability to register more than one device, keep calibration histories separate, process messages consistently and version model outputs. A central inference service could aggregate records from several devices while ensuring that samples are not accidentally mixed. Model retraining should be evaluated across hardware revisions, not just the original sensor board. Firmware updates and credentials need a documented maintenance lifecycle.

Ownership of project code, data and model artefacts must be reconciled with SLIIT policy, third-party licenses and the contributions of other team members. Public deployment would additionally require an assessment of applicable electrical safety,

wireless communication, data protection and water-testing claims. The thesis cannot infer regulatory approval from a university prototype.

8.5 Contribution to the Sustainable Development Goals

The research concept is relevant to Sustainable Development Goal 6, clean water and sanitation, because monitoring information can support attention to water resources. It also has potential connections to responsible innovation and education. These are alignments of purpose rather than demonstrated impact. Actual SDG contribution would require measures such as access improvement, verified risk reduction or demonstrable operational value in a defined population.

A sustainability assessment should account for electronics lifetimes, sensor replacement, calibration solutions, power requirements and e-waste. Long-term practicality depends as much on maintainability and interpretation as on a low purchase price. An inexpensive instrument that produces misleading labels could undermine rather than support sustainable water management.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis draft establishes a technically coherent, source-grounded framework for the individual research component “Different Types of Water Identifier – ML & AI” within the Water Purification project. It clarifies the classification task, its distinction from drinking-water safety assessment, relevant measurement principles, proposed model comparisons, acquisition protocol, software architecture and testable performance criteria. The framework provides a route to demonstrating a practical low-cost sensor-ML system with greater evidential integrity than a classifier trained and scored on ungrouped time-series rows.

The uploaded materials do not document the final implemented system, measured accuracy, runtime latency or user-study outcomes. Consequently, it would be premature to conclude that the research objectives have been empirically achieved. The strongest defensible conclusion at this stage is that a reproducible solution and validation plan have been articulated from the original proposal. The final thesis must be revised using author-supplied experimental and software evidence before certification or submission.

9.2 Achievement of Objectives

Objective achievement requires mapped evidence. Sensor acquisition is demonstrated by wiring records, calibration logs and packet captures; dataset preparation by the recorded sample manifest and cleaned table; algorithm comparison by reproducible cross-validation and held-out predictions; integration by service and dashboard test logs; and system validation by independent physical samples and operational metrics. The supplied files establish that these objectives were proposed, but they are insufficient to score them as completed.

The numerical targets of 90% accuracy and less than two seconds response time remain unverified. The indicative budget provides a starting estimate, not an invoice-supported claim. The final version should include a concise achieved/not-achieved/partially-achieved table with evidence links and explanations of any departures from the proposed design.

9.3 Individual Research and Product Contributions

The intended individual research output is a transparent multi-parameter classification methodology with defensible labels, feature preprocessing, comparative algorithms, sample-aware evaluation and a clearly scoped output interface. The product output is the proposed water identifier module that can expose measurement data, class and quality status to a larger group system. Actual ownership must be confirmed using repository contributions, model files and test evidence.

The most important contribution to professional practice is a safety-aware interpretation boundary. A model prediction can assist classification tasks but cannot establish potable-water status from a handful of proxy measurements. Keeping that boundary explicit makes the resulting research more reliable, even if eventual accuracy is lower than initially expected.

9.4 Future Directions

The immediate priority is to complete an independently labelled dataset covering several physical sources per class and to freeze a leakage-resistant evaluation split. Next priorities are instrument calibration and cross-device checks, execution of the baseline/model comparison, reproducible uncertainty reporting and confirmation of end-to-end response times. Robustness should be challenged through temperature changes, normal sensor drift, unfamiliar mineral profiles and borderline class examples.

Further extensions may include independent ion-specific measurements, explicit novelty detection for unknown samples, calibrated prediction confidence, on-device inference and an auditable integration service for downstream group components. Automated treatment decisions would require a separate safeguarded validation programme and approval against relevant public-health or engineering standards. User-study feedback, sustained operating-cost analysis and better market evidence would strengthen commercial feasibility.

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GLOSSARY

- **Calibration** — Comparison against a traceable reference to estimate and correct instrument response.
- **Class label** — A category used as the supervised target for training and evaluation.
- **Concept drift** — A change over time in the relationship between sensor inputs and target classes.
- **Data leakage** — Use of information during training that would not be available for truly independent future predictions.
- **Electrical conductivity** — A measure of how readily water carries current; influenced by dissolved ions and temperature.
- **Ground truth** — Reference information used to assign target classes; its collection method must be documented.
- **Macro-F1** — Unweighted mean of per-class F1 values, useful when class frequencies differ.
- **Out-of-distribution sample** — An input different from the samples on which the model was trained and evaluated.
- **Sample-aware split** — Division of dataset by physical sample or batch, not by individual repeated sensor row.
- **Turbidity** — Optical scattering of light by suspended particles, typically reported in NTU.
- **Water type** — A task-dependent label such as distilled, tap, mineral or alkaline, not a universal safety judgement.

APPENDICES

Appendix A : Research instruments or extended results

The following forms are proposed research artefacts. Empty fields are intentionally retained because the original proposal did not provide verified sensor logs, training outputs, model scores or approved stakeholder responses. The candidate should complete these records with dated source evidence before final submission.

Table A.1: Physical sample registration form

Field	Required entry / example format
Sample ID	Unique pseudonymous code
Water-type reference label	One of pre-registered class labels
Reference method	Manufacturer documentation / laboratory analysis / controlled preparation
Source batch and sample container	Unique coded provenance
Sampling date/time	ISO 8601 and local timezone
Collection and storage conditions	Temperature, exposure duration, rinsing
Device and calibration ID	Versioned hardware references
Operator and anomalies	Initials, notes without sensitive information

Table A.2: Calibration and equipment record

Parameter	Record required
pH	Buffer values, observed values, slope/offset, calibration date
TDS / EC	Reference solution, conversion factor, temperature compensation
Turbidity	Standard and measurement protocol, zero check
Temperature	Reference comparison and sensor offset
Firmware	Commit/version and acquisition frequency
Device wiring	Pinout, power levels and protective components

Table A.3: Algorithm comparison result sheet

Model	Macro-F1	Accuracy	p95 latency	Evidence ID
Majority-class baseline	Pending	Pending	Pending	—
Logistic regression	Pending	Pending	Pending	—
KNN	Pending	Pending	Pending	—
SVM	Pending	Pending	Pending	—
Random forest	Pending	Pending	Pending	—
Gradient boosting	Pending	Pending	Pending	—

Table A.4: Objective-to-evidence traceability matrix

Objective	Documentary evidence needed	Status
Hardware and acquisition	Circuit, sensor list, serial records, calibration	Awaiting
Labelled dataset	Source batch log, raw CSV, data dictionary	Awaiting
Algorithm study	Code, split, confusion matrix, class metrics	Awaiting
Live inference	API trace, model bundle and dashboard video/screens	Awaiting
User interface	Accessibility check and task feedback	Awaiting
Independent validation	Blind samples, reference labels and test report	Awaiting

Recommended raw evidence retention: export sensor packets in an immutable dated directory; generate SHA-256 hashes for raw logs; retain the data-split manifest; preserve project repositories or versioned archives; include model checkpoints, environment lockfiles, test execution output and representative screenshots with redacted personal details. These items are not provided in this draft.

Appendix B : Academic integrity and AI use disclosure

This thesis draft was prepared with generative-AI assistance for organisation, academic phrasing, design schematics and document creation. The author must review all text, ensure the work accurately represents their own contributions, verify all sources and comply with the applicable SLIIT module disclosure rules. AI assistance

does not establish experimental facts, replace original research or transfer responsibility for errors from the author.

Proposed final disclosure fields: tool name and version; purpose; representative prompts or workflow description; chapters affected; confirmation of independent source verification; confirmation that no unobserved accuracy or experimental results were fabricated; author name; reviewer signature and date. Each field must be completed truthfully for the final academic submission.

Appendix C : Proposal-to-thesis traceability and unresolved evidence

The source proposal, “Water Purification (Different Types of Water Identifier – ML & AI),” was used as the authoritative project statement. Its sensors and class labels inform Chapters 1, 3 and 4; algorithm candidates inform Chapters 2–5; numeric targets appear as proposed requirements, not outcomes; the historical budget appears in Chapter 8. The friend’s individual report was consulted only to understand presentation expectations and no research claims or copied analysis were imported.

Before finalisation, confirm the list of features that the candidate says have been added, removed or altered since March 2026; identify final hardware, code, trained model, performance logs and screenshots; confirm the actual submission month; and reconcile the final student name, supervisor certification and authorship disclosure. The current draft deliberately does not silently substitute guesses for this missing information.

Appendix D : Operational acceptance and safety checklist

- All five inputs have explicit units and calibration references.
- Physical sample IDs remain separated across final training and test partitions.
- Selected hyperparameters and preprocessor are fitted without test-data leakage.
- Unknown, malformed and stale readings produce explicit non-deceptive states.
- The UI does not state that an ML water-type prediction certifies drinking-water safety.
- All claim-supporting figures and metrics can be reproduced from committed files.
- Actual hardware invoices reconcile with the initial proposed budget.
- The supervisor and student have reviewed the reference list, declarations and AI disclosure.

Appendix E : Original proposal methodology and architecture figures

The following two diagrams reproduce the original proposal graphics as supplied by the author. They document the March 2026 proposed design and should not be interpreted as evidence that the hardware, machine-learning models or user interface were implemented or evaluated. Some labels and hardware choices in the drawings may differ from the final implemented system; the candidate must reconcile them against source artefacts.